

SAFETY DATA SHEET (GHS)

Ethylene Glycol (and 80wt% Water / 20wt% Ethylene Glycol heat-transfer mixture) | CAS: 107-21-1 | Formula: C₂H₆O₂

PTT Phenol Train II (PPCL), Map Ta Phut, Thailand — HAZOP PSI Document | Issued: 2026-06-13 | SDS Version 1.0

DATA SOURCES CONSULTED

- ECHA Classification & Labelling Inventory — Ethylene glycol, CAS 107-21-1 (<https://echa.europa.eu/>)
- PubChem CID 174 — GHS Classification (Rev.9), <https://pubchem.ncbi.nlm.nih.gov/compound/174>
- ILO/NIOSH International Chemical Safety Card (ICSC) 0270 — Ethylene glycol
- ACGIH TLV/BEI — Ethylene glycol (Ceiling, aerosol fraction)
- NOAA CAMEO Chemicals — Ethylene glycol datasheet (<https://cameochemicals.noaa.gov/>)
- Accessed 2026-06-13

SECTION 1 — IDENTIFICATION

Product identifier: Ethylene Glycol (pure); also supplied/used on-site as an 80wt% Water / 20wt% Ethylene Glycol heat-transfer (anti-freeze/coolant) mixture

CAS Number: 107-21-1 (ethylene glycol component); water CAS 7732-18-5

Chemical formula: C₂H₆O₂ (ethylene glycol); mixture = H₂O + C₂H₆O₂

Molecular weight: 62.07 g/mol (ethylene glycol)

Synonyms: 1,2-Ethanediol; Glycol; Monoethylene glycol (MEG)

Intended use: Closed-loop heat-transfer/coolant fluid (anti-freeze) for equipment cooling/winterisation circuits. PTT Phenol Train II (PPCL), Map Ta Phut, Thailand

Emergency telephone: [Plant emergency response number — to be inserted by PPCL HSE]

SECTION 2 — HAZARD IDENTIFICATION

GHS classification (pure ethylene glycol): Acute Tox. 4 (H302, oral); STOT RE 2 (H373 — kidneys, via repeated oral/significant exposure)

Signal word: Warning

Hazard statements (H-codes): H302 (Harmful if swallowed); H373 (May cause damage to organs [kidneys] through prolonged or repeated exposure)

Precautionary statements (P-codes): P260, P264, P270, P301+P312, P314, P330, P501

GHS pictograms: GHS07 (Exclamation mark)

80/20 water/glycol mixture classification: At 20wt% ethylene glycol, the mixture falls below the generic concentration limit typically used for classification of EG-containing mixtures for Acute Tox. 4 (H302) under additivity-based mixture classification rules in many jurisdictions (commonly ~10-25% cut-off range depending on the classification scheme applied) — CONFIRM the mixture-specific classification against the supplier SDS for the actual 80/20 product used on-site; this SDS presents the hazard profile of the pure ethylene glycol component as the conservative basis pending that confirmation.

Other hazards: Low acute toxicity by skin contact and low volatility (high boiling point, low vapour pressure) mean inhalation/dermal hazards are limited under normal handling. Principal hazards are: (1) ingestion toxicity (sweet taste creates an attractant/ingestion hazard, particularly relevant for animals and for accidental ingestion), and (2) slip hazard from spills (viscous liquid). Not flammable under normal ambient conditions (high flash point, ~111°C) but combustible at elevated temperature — relevant if used near hot equipment.

SECTION 3 — COMPOSITION / INFORMATION ON INGREDIENTS

Ethylene glycol (pure feedstock/make-up): CAS 107-21-1 — typically >99% purity as supplied

80/20 Water/Glycol heat-transfer mixture — composition: Water (CAS 7732-18-5): 80% w/w; Ethylene glycol (CAS 107-21-1): 20% w/w. This composition is the as-used heat-transfer fluid concentration referenced in the HAZOP PSI brief for Train II cooling/winterisation circuits.

Mixture additives: Commercial heat-transfer formulations typically include corrosion inhibitors (e.g. borate, nitrite, or organic acid salt-based inhibitor packages) and may include a dye — confirm the specific inhibitor package used on-site

against the supplier SDS, as some inhibitors carry their own hazard classifications (e.g. amine-based inhibitors)

Note on physical property data below: Sections 9-10 below primarily describe PURE ethylene glycol. The 80/20 water/glycol mixture will have physical properties intermediate between water and pure EG (e.g. freezing point depressed to approximately -8 to -10°C for a 20wt% EG/water mixture, boiling point close to that of water, viscosity and density close to water with a modest increase) — obtain the supplier's mixture-specific data sheet for precise values used in heat-transfer/freeze-protection design calculations.

SECTION 4 — FIRST AID MEASURES

Inhalation: Under normal conditions (low vapour pressure), inhalation exposure is unlikely except from heated mist/aerosol. Remove to fresh air if aerosol/mist exposure occurs. Obtain medical attention if symptoms develop.

Skin contact: Wash with soap and water. Remove contaminated clothing. Low acute dermal toxicity — primarily a cleanliness/slip-hazard consideration.

Eye contact: Flush with water for several minutes. Obtain medical attention if irritation persists.

Ingestion: Do NOT induce vomiting. Obtain immediate medical attention even if the person appears well — ethylene glycol ingestion can cause delayed, severe metabolic acidosis and acute kidney injury (H373) due to toxic metabolites (glycolic acid, oxalic acid); early medical intervention (e.g. fomepizole or ethanol therapy, haemodialysis) is time-critical and significantly improves outcomes.

Notes to physician: Ethylene glycol poisoning: monitor osmolar and anion gap, blood ethylene glycol level, and renal function; calcium oxalate crystalluria is a classic finding; fomepizole (4-methylpyrazole) is the preferred antidote to block alcohol dehydrogenase-mediated metabolism to toxic metabolites; haemodialysis may be required in severe poisoning

SECTION 5 — FIREFIGHTING MEASURES

Suitable extinguishing media: Water spray/fog, dry chemical, CO₂, alcohol-resistant foam — appropriate for surrounding fire as ethylene glycol itself has a high flash point and is combustible rather than highly flammable

Unsuitable extinguishing media: None specifically identified

Special hazards: Combustible liquid at elevated temperature (flash point ~111°C); not a significant fire hazard under normal ambient conditions, but heated equipment/leaks onto hot surfaces could ignite. Heated/burning ethylene glycol can produce irritating fumes.

Special protective equipment for firefighters: Standard structural firefighting PPE with SCBA for fires involving heated EG or surrounding combustibles

Hazardous combustion products: Carbon monoxide, carbon dioxide, and potentially acetaldehyde/other aldehydes from incomplete combustion

SECTION 6 — ACCIDENTAL RELEASE MEASURES

Personal precautions: Low acute hazard for short-duration cleanup; avoid prolonged skin contact and ingestion. Be aware of slip hazard from spilled liquid (viscous, can be mistaken for water).

Environmental precautions: Prevent entry to drains/watercourses in large quantities — while individually low in aquatic toxicity, EG exerts a high biochemical oxygen demand (BOD) and can cause oxygen depletion in receiving waters if released in bulk

Containment and cleanup: Absorb with inert absorbent material; collect for disposal/recycling. Wash residual area with water. Be alert to the attractant (sweet-tasting) nature of EG spills with respect to animals/wildlife in the vicinity.

Reference to other sections: See Section 7 (Handling/Storage) and Section 13 (Disposal)

SECTION 7 — HANDLING AND STORAGE

Handling precautions: Standard precautions for a low-volatility, combustible liquid: avoid ingestion, prolonged skin contact; minimise misting/aerosol generation

Storage conditions: Store in closed containers/tanks in a well-ventilated area away from strong oxidisers and incompatible materials; protect from contamination of potable water systems

Temperature limits: Store within manufacturer's recommended temperature range; for the 80/20 water/glycol mixture, confirm the freeze-protection temperature (~-8 to -10°C for 20wt% EG) is adequate for the intended service temperature range — heat tracing/insulation may still be required for extreme cold events depending on design margins

Pressure: Atmospheric storage; closed-loop circuits operate per system design pressure

Segregation requirements: Segregate from strong oxidisers and food/drinking water systems (cross-connection control — EG contamination of potable water is a recognised hazard at industrial sites)

Special requirements: Backflow prevention/cross-connection control between glycol-based heat-transfer loops and potable water systems; secondary containment for bulk EG storage

Specific end-use recommendations: Maintain the 80/20 water/glycol ratio per freeze-protection design requirements; periodically test inhibitor concentration and pH of in-service glycol fluid per manufacturer's maintenance guidance, as degraded inhibitor packages can lead to corrosion of the cooling circuit

SECTION 8 — EXPOSURE CONTROLS / PERSONAL PROTECTION

Occupational Exposure Limits (OEL): ACGIH TLV-Ceiling: 100 mg/m³ (aerosol fraction only — vapour exposure is negligible under normal conditions due to low volatility). No OSHA PEL established for ethylene glycol vapour. Thailand OEL: confirm current Ministry of Labour value if any.

TWA: Not established for vapour (low volatility); see Ceiling value for aerosol

STEL: Not established

Ceiling: 100 mg/m³ (ACGIH, aerosol/particulate fraction)

Skin notation: Not assigned

IARC/NTP/OSHA carcinogen status: Not classified as a carcinogen

Engineering controls: General ventilation adequate for normal handling; local exhaust if heated misting/aerosol-generating operations occur

PPE — Respiratory: Not normally required; use if aerosol/mist generation occurs (e.g. heated spray applications)

PPE — Gloves: Chemical-resistant gloves (nitrile or similar) for routine handling

PPE — Eyes: Safety glasses; goggles for tasks with splash potential

PPE — Body: Standard work clothing; chemical-resistant apron for bulk transfer tasks

Biological monitoring: Not routinely applicable for occupational exposure at expected levels; clinical biomarkers (serum/urine EG, anion/osmolar gap) are used only in suspected acute poisoning

SECTION 9 — PHYSICAL AND CHEMICAL PROPERTIES

Appearance: Colourless, odourless, viscous, hygroscopic liquid (pure EG); 80/20 mixture appears as a slightly viscous, water-like liquid, often dyed by manufacturer for identification

Odour: Practically odourless (pure EG); slight sweet odour possible

Odour threshold: Not well established (low volatility)

pH: Pure EG ~6-7.5 (neutral); in-service heat-transfer fluid pH depends on inhibitor package — typically maintained alkaline (pH ~8-10) by the inhibitor

Melting point / Freezing point: -12.9°C (pure EG); 80/20 water/glycol mixture approx. -8 to -10°C (confirm against supplier freeze-point chart for exact composition)

Boiling point (at 1 atm): 197.3°C (pure EG); 80/20 mixture boils close to water's boiling point (slightly elevated, ~101-103°C) due to the low EG fraction

Flash point: 111°C (closed cup, pure EG) — experimental literature value; the 80/20 water/glycol mixture has no meaningful flash point under normal conditions due to the high water content

Evaporation rate: Very low (pure EG: <0.01 relative to n-butyl acetate)

Flammability (solid/gas): Pure EG: combustible liquid (not 'highly flammable'); 80/20 mixture: not flammable under normal conditions

Flammability limits in air: LEL ~3.2 vol%; UEL ~53 vol% (pure EG vapour, at elevated temperature — not relevant at ambient temperatures given low volatility)

Vapour pressure (at 20°C): ~0.06 hPa (pure EG) — negligible

Vapour density (air = 1): ~2.1 (pure EG vapour, where present)

Relative density (water = 1): ~1.11-1.115 (pure EG at 20°C); 80/20 mixture ~1.02-1.03

Solubility (water): Fully miscible (pure EG); mixture is by definition a homogeneous water solution

Partition coefficient (log K_{ow}): ~-1.36 (pure EG, hydrophilic)

Auto-ignition temperature: ~410°C (pure EG)

Decomposition temperature: Not applicable under normal use

Viscosity: ~16-21 mPa·s at 25°C (pure EG, notably more viscous than water); 80/20 mixture only slightly more viscous than water at the same temperature

Explosive properties: Not classified as explosive

Oxidising properties: Not an oxidiser

Molecular weight: 62.07 g/mol (pure EG)

SECTION 10 — STABILITY AND REACTIVITY

Reactivity: Stable under normal conditions; can react with strong oxidisers (potentially exothermically) and with strong acids (esterification)

Chemical stability: Stable under recommended storage conditions; in-service heat-transfer fluid can degrade thermally/oxidatively over time, forming acidic degradation products (e.g. glycolic acid) if inhibitor depletion occurs — periodic fluid testing/replacement is part of normal maintenance

Possibility of hazardous reactions: Can react vigorously with strong oxidising agents (e.g. chromic acid, permanganates, nitric acid) — exothermic, potentially hazardous reaction; thermal decomposition at high temperature (e.g. fire exposure or severe overheating in a heat-transfer circuit) can produce irritating/toxic fumes including aldehydes (e.g. acetaldehyde) and carbon monoxide

Conditions to avoid: Contact with strong oxidisers; prolonged overheating beyond manufacturer-recommended fluid temperature limits (accelerates degradation and corrosive by-product formation)

Incompatible materials: Strong oxidisers, strong acids (esterification), strong bases (in concentrated form)

Hazardous decomposition products: Thermal decomposition/combustion can produce carbon monoxide, carbon dioxide, and aldehydes (e.g. acetaldehyde, formaldehyde) — relevant if heat-transfer fluid is exposed to severe overheating (e.g. localised hot spots/fouling in heat exchangers)

SECTION 11 — TOXICOLOGICAL INFORMATION

Routes of exposure: Ingestion (primary concern — H302/H373), dermal/ocular contact (low hazard), inhalation (low hazard at normal temperatures)

Acute toxicity — Oral LD50: ~4,700-13,000 mg/kg, rat (variable by source; classified Acute Tox. 4 — Harmful if swallowed, H302)

Acute toxicity — Dermal LD50: >10,000 mg/kg (low acute dermal toxicity — not classified)

Acute toxicity — Inhalation LC50: Not readily applicable due to negligible vapour pressure under normal conditions; not classified for acute inhalation toxicity

Skin corrosion/irritation: Not classified — may cause mild irritation on prolonged contact

Serious eye damage/irritation: Not classified as a serious eye hazard; may cause mild transient irritation

Respiratory/skin sensitisation: Not classified as a sensitiser

Germ cell mutagenicity: Not classified

Carcinogenicity: Not classified as a carcinogen

Reproductive toxicity: Some developmental effects have been reported in animal studies at very high doses, but ethylene glycol is not formally classified as a reproductive toxicant under the harmonised GHS classification — confirm current ECHA status if relevant to occupational risk assessment

STOT (single exposure): Not classified for single-exposure STOT (note: ingestion toxicity, H302/H373, is the dominant hazard rather than inhalation/dermal STOT)

STOT (repeated exposure): May cause damage to organs (kidneys) through prolonged or repeated exposure (H373) — primarily relevant to repeated/chronic ingestion exposure scenarios rather than typical occupational inhalation/dermal exposure

Aspiration hazard: Not classified

SECTION 12 — ECOLOGICAL INFORMATION

Aquatic toxicity (LC50/EC50): Low acute aquatic toxicity — fish LC50 (96h) typically >1,000-10,000 mg/L; not classified for acute/chronic aquatic hazards under GHS

Persistence and degradability: Readily biodegradable; however, exerts a high biochemical oxygen demand (BOD) — large releases to water bodies can cause significant oxygen depletion and fish kills via indirect (BOD) mechanism rather than direct toxicity

Bioaccumulative potential: Negligible (log K_{ow} ~-1.36)

Mobility in soil: Highly mobile (fully water miscible)

Other adverse effects: Spent/degraded glycol heat-transfer fluids may contain metal corrosion products and degraded inhibitor compounds — assess as a potentially distinct waste stream from virgin EG

SECTION 13 — DISPOSAL CONSIDERATIONS

Disposal methods: Spent glycol fluid should be assessed for contamination (degradation products, metals from corrosion) before disposal; recycling/reclamation via specialist glycol recovery contractors is preferred where available; otherwise dispose per local hazardous/non-hazardous waste classification and site procedures

Contaminated packaging: Rinse with water; dispose of per local regulations

Regulatory reference: Thailand Hazardous Substance Act B.E. 2535 (verify classification status of EG and spent glycol fluid); site effluent discharge permit (BOD/COD limits)

SECTION 14 — TRANSPORT INFORMATION

UN Number: Not regulated as a dangerous good for transport (pure ethylene glycol is generally not assigned a UN number under standard transport regulations); confirm classification of the specific commercial heat-transfer fluid product (inhibitor package may affect classification)

Proper shipping name: Not applicable / Not regulated (confirm for specific product)

Transport hazard class(es): Not classified as a transport hazard (pure EG)

Packing group: Not applicable

Environmental hazards (marine pollutant): Not classified as a marine pollutant

Special precautions for user: None specific beyond standard good practice for bulk liquid transfer

Bulk transport (MARPOL/IBC): Not typically subject to IBC Code special provisions; confirm for bulk marine shipment if applicable

SECTION 15 — REGULATORY INFORMATION

Safety and health regulations: See country-specific notes below

Thailand: Hazardous Substance Act B.E. 2535 — classification depends on concentration/use; verify listing (List 1-4) with Department of Industrial Works.

EU: Registered under REACH (Regulation (EC) No 1907/2006); classified under CLP Regulation (EC) No 1272/2008.

US: Subject to OSHA Hazard Communication Standard 29 CFR 1910.1200 (HazCom 2012, aligned with GHS Rev.7).

ASEAN: Classification consistent with ASEAN GHS implementation (harmonised with UN GHS).

Chemical safety assessment: Yes — performed by upstream registrants under REACH; site-specific CSA recommended as part of PPCL PSI documentation.

SECTION 16 — OTHER INFORMATION

Revision date: 2026-06-13

Revision number: 1.0

Key changes from previous version: Initial issue — compiled for HAZOP PSI (CAS 107-21-1), including the 80wt% water / 20wt% ethylene glycol heat-transfer mixture composition documented in Section 3 per HAZOP PSI requirements.

Data sources used in this SDS: See 'Data Sources Consulted' block at top of document

Disclaimer: This SDS was compiled for HAZOP Process Safety Information purposes at PTT Phenol Train II (PPCL), Map Ta Phut, Thailand. Data sourced from published authoritative references (ECHA, PubChem/GHS Rev.9, NIOSH, NTP, supplier SDSs, CAMEO Chemicals, peer-reviewed literature). Verify against current manufacturer SDS before use for regulatory compliance or labelling purposes.

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